MALAK SOBHY ABDELATTY

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PROFESSIONAL SUMMARY

AI & Machine Learning Engineer with hands-on experience in developing machine learning and deep learning models, building LLM-based applications, and designing secure AI systems. Skilled in PyTorch, TensorFlow, and Scikit-learn, with experience in Retrieval-Augmented Generation (RAG), model evaluation, Federated Learning, adversarial machine learning, and privacy-preserving AI.

EDUCATION

Bachelor of Computer Science | Cairo University *Sep 2022 - June 2026*
Rank: 4th in Cohort

CERTIFICATIONS

AI Security Mastery: From Attacks to Defenses *September 2026*
MaharaTech (ITI), Artificial Intelligence Academy – Ministry of Communications and Information Technology. Hands-on projects in adversarial attacks, data poisoning, and privacy-preserving AI.

SKILLS

AI & Machine Learning: PyTorch, TensorFlow, Keras, Scikit-learn, CNNs, LSTMs, Exploratory Data Analysis (EDA), Model Evaluation.

LLM & AI Application Development: LangChain, Retrieval-Augmented Generation (RAG), Embeddings, Vector Search, ChromaDB, LLM Evaluation.

AI Security & Privacy: Adversarial Machine Learning, FGSM, PGD, Data Poisoning, Federated Learning, Privacy-Preserving ML, Homomorphic Encryption (FHE), Encrypted Inference.

Libraries & Tools: PySyft, Concrete-ML, TenSEAL, Adversarial Robustness Toolbox (ART), FastAPI, Pydantic, SlowAPI, Redis, Docker, Git.

Programming & Databases: Python, SQL, SQLite, Java, JavaScript, Node.js, MongoDB, Firebase, Linux, Angular, React Native.

PROJECTS

3ala Damanty: Evaluator–Generator RAG Platform *Oct 2026*

- Developed a Retrieval-Augmented Generation (RAG) platform that ingests multiple document formats, web pages, Wikipedia content, and audio transcriptions.
- Designed a Generator–Evaluator feedback loop that iteratively evaluates and refines answers to improve quality, with up to four refinement attempts.
- Integrated vector-based retrieval with ChromaDB to ground LLM responses in relevant source content.
- Implemented conversation memory and persistent storage (SQLite, Redis) for generated answers and evaluation history.

Skills: *Python, LangChain, RAG, ChromaDB, SQLite, Redis, LLM Evaluation*

Adversarial Attacks & Defenses *Sep 2026*

- Implemented FGSM and PGD adversarial attacks to evaluate the robustness of an image classification model. Performed epsilon sensitivity analysis and implemented adversarial training to improve model resilience against adversarial examples.

Skills: *Python, PyTorch, Adversarial ML, FGSM, PGD, Adversarial Training, ART*

Data Poisoning Attacks & Defenses *Aug 2026*

- Implemented Dirty-Label and Backdoor Poisoning attacks against a credit-card fraud detection model. Applied Isolation Forest as an anomaly-based defense and evaluated attack effectiveness using Fraud Recall, PR-AUC, and Attack Success Rate.

Skills: *Python, Scikit-learn, Data Poisoning, Backdoor Attack, Dirty-Label Attack, Isolation Forest, ML Security*

SecureSentinel AI API *Aug 2026*

- Built and deployed a privacy-first sentiment analysis API that detects and anonymizes Personally Identifiable Information before inference. Implemented asynchronous REST endpoints with FastAPI, Pydantic validation, SlowAPI rate limiting, Docker containerization, and cloud deployment.

Skills: *Python, FastAPI, SlowAPI, Docker, Cloud Deployment*

Federated Heart Disease Classification *July 2026*

- Implemented a Federated Learning pipeline simulating two hospital nodes where patient data remains local. Trained models locally and aggregated model parameters using FedAvg. Refactored legacy PySyft code to work with a modern PySyft release using @sy.syft_function, achieving 84.47% accuracy on unseen holdout data.

Skills: *Python, PyTorch, PySyft, Federated Learning, FedAvg, Privacy-Preserving ML, Remote Execution*

Ironwall - Privacy-Preserving Network Security *June 2026*

- Developed a privacy-preserving ML pipeline for network security that performs inference directly on encrypted traffic using Fully Homomorphic Encryption. Built and evaluated FHE-compatible models with Concrete-ML, replacing ReLU with quadratic activation and removing Batch Normalization to ensure encrypted computation stability. Optimized inference through simulation mode, reducing per-record execution time from minutes to seconds.

Skills: *Python*, *Machine Learning*, *FHE*, *Concrete-ML*, *TenSEAL*, *PPML*, *Encrypted Inference*, *Network Security*

EXPERIENCE

Machine Learning & Data Analysis Trainee | NTI *Oct 2025 – Feb 2026*

- Implemented 5+ predictive models using Scikit-Learn and TensorFlow, improving baseline accuracy by 12% through EDA and advanced feature engineering.
- Enhanced Deep Learning Frameworks workflows, reducing training error by 15% through rigorous hyperparameter tuning and cross-validation techniques.

MEAN Stack Intern | NTI *Aug 2025 – Sep 2025*

- Engineered and deployed full-stack web applications using MongoDB, Express.js, Angular, and Node.js.

COMPETITIONS & ACTIVITIES

Egyptian Collegiate Programming Contest (ECPC) *2024*

- Solved complex algorithmic problems using Java, demonstrating strong data structure and logic proficiency.